

Machine learning in molecular dynamics beyond potentials

Filippo Bigi
EPFL

filippo.bigi@epfl.ch

Simulation I: conservative forces



The neural network is a function

$$V(q)$$



Differentiate to get forces

$$F(q) = -dV(q)/dq$$

(expensive)



Run MD step

Simulation II: non-conservative forces

The neural network is a function

~~$V(q)$~~ $F(q)$



Run MD step

Skips expensive
differentiation...

Simulation II: non-conservative forces

The neural network is a function

~~$V(q)$~~ $F(q)$



Run MD step

Skips expensive
differentiation...

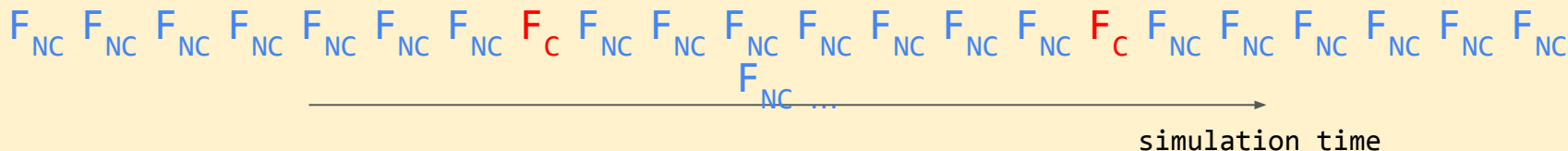
...but the total energy is not conserved anymore.

Simulation III: multiple time stepping

Multiple time stepping (MTS) restores energy conservation

$$F_c(q) = -dV(q)/dq$$

$$F_{NC}(q) \text{ (direct, fast)}$$



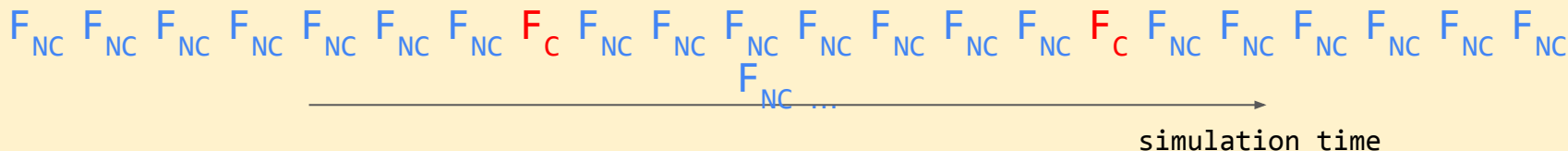
Fast and yields observables consistent with fully conservative simulations

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Fast and yields observables consistent with fully conservative simulations

```
pair_style hybrid/overlay &  
  metatomic_1 pet-mad-s-v1.5.0.pt non_conservative on scale 1.0 &  
  metatomic_2 pet-mad-s-v1.5.0.pt non_conservative on scale -1.0 &  
  metatomic_3 pet-mad-s-v1.5.0.pt non_conservative off scale 1.0  
run_style respa 2 8 hybrid 1 2 2
```

Simulation IV: FlashMD

FlashMD learns and then predicts large steps

Step 1 → Step 2 → Step 3 → Step 4 → Step 5 → Step 6



FlashMD

Conservative throughput: 0.028 ns/day

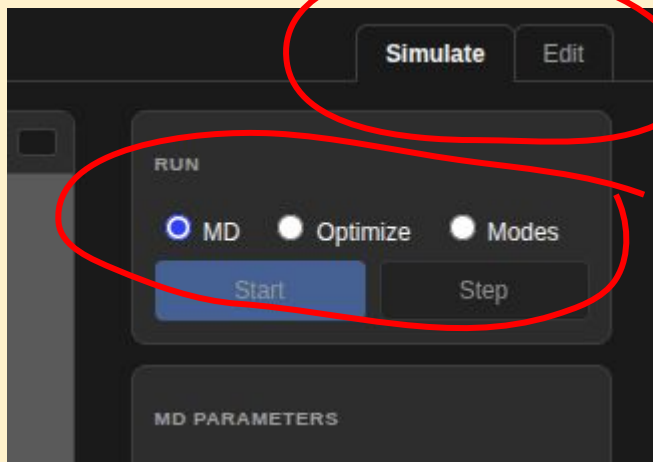
Non-conservative throughput: 0.056 ns/day

FlashMD throughput: 0.811 ns/day

```
fix flashmd all metatonic flashmd-16.pt types 8 1
```

Interactive MD - Part I: mlip.cpp

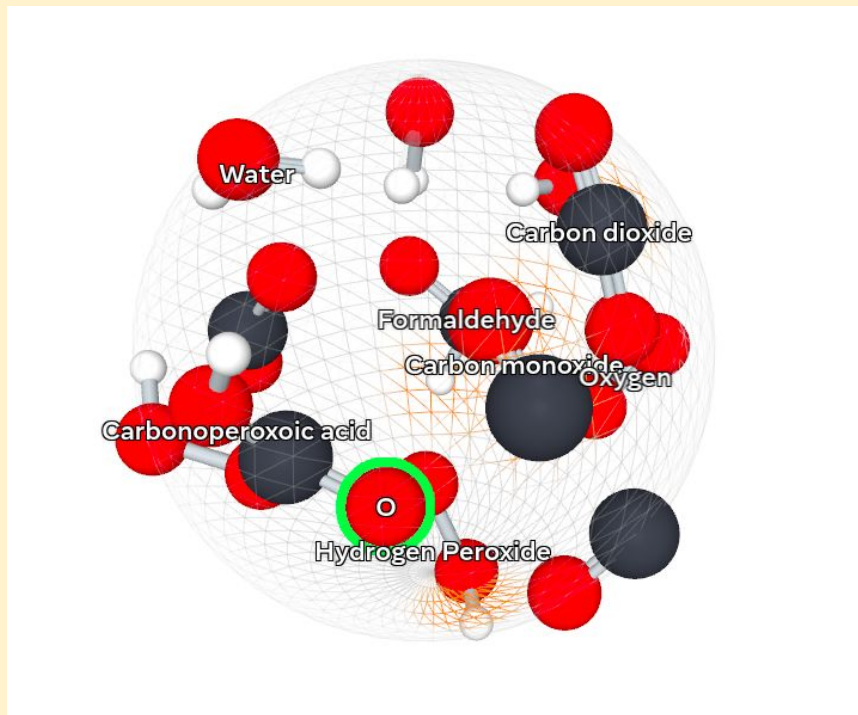
Light model, runs in your browser, add atoms, conservative and non-conservative options, also vibrations/phonons



“fast” = non-conservative

Interactive MD – Part II: UMA demo

Heavy model, run remotely, cool tutorials



Interactive MD – Part II: UMA demo

Heavy model, run remotely, cool tutorials

Meta's Universal Model for Atoms (UMA) Demo

Model version: UMA 1.2 — [see changelog](#)

- 1. UMA Intro
- 2. Explore UMA's capabilities
- 3. Try UMA on your structures!

As the final step of the demo, try running your own structure through the UMA model!

To use a custom input structure with this demo:

1. [Request gated model access](#). Requests for model access are typically processed within a matter of minutes.
2. Login to Hugging Face using the "Sign in with Hugging Face button" in the inputs section.
3. Then upload a structure file below and click run!
 - Note that uploaded structure will be stored by this demo to analyze model usage and identify domains where model accuracy can be improved.

Interactive MD – Part III: MLIP playground

Model Selection

Select Model Type:

- ☐ MACE
- ☐ FairChem
- ☐ ORB
- ☐ SEVEN_NET
- ☐ MatterSim
- ☒ UPET
- ☐ UFF
- ☐ D3 dispersion
- ☐ xTB

Select UPET Model:

PET-MAD-XS-V1.5.0

- ☒ Direct (non-conservative forces)?
- ☐ Uncertainty Prediction?

Results

Energy: -15.290419 eV

Time Taken: 0.2446 seconds

Energy Uncertainty: 0.125537 eV

Energy Ensemble: [-15.256793 -15.135869 -15.364841 -15.198721 -15.297085 -15.229199 -15.253027 -15.293375 -15.301746 -15.355101 -15.304639 -15.301284 -15.295305 -15.356879 -15.3210745 -15.316568 -15.226757 -15.290802 -15.22846 -15.276125 -15.280624 -15.254757 -15.302967 -15.390768 -15.231206 -15.264715 -15.301431 -15.223002 -15.283356 -15.314323 -15.308483 -15.258551 -15.356678 -15.272554 -15.282472 -15.215313 -15.297269 -15.300629 -15.270398 -15.31805 -15.263241 -15.20934 -15.305558 -15.219164 -15.308375 -15.350207 -15.335778 -15.326472 -15.300204 -15.1939 -15.412304]